

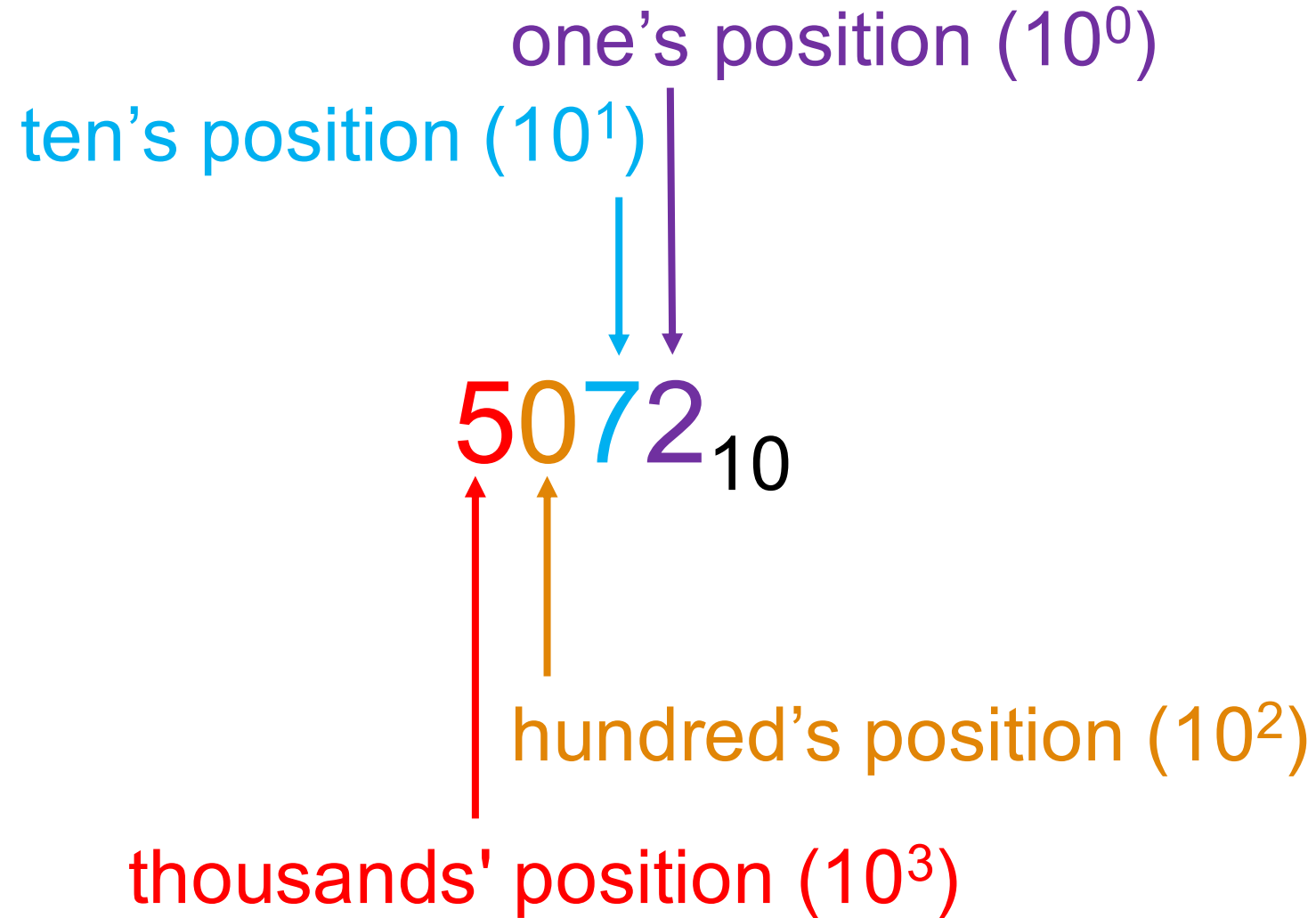
A decorative graphic on the left side of the slide, consisting of a network of thin, gold-colored lines and small circles, resembling a circuit board or a neural network diagram.

Number Representation

COMP211 Connor McMahon

Base-10

Decimal Notation



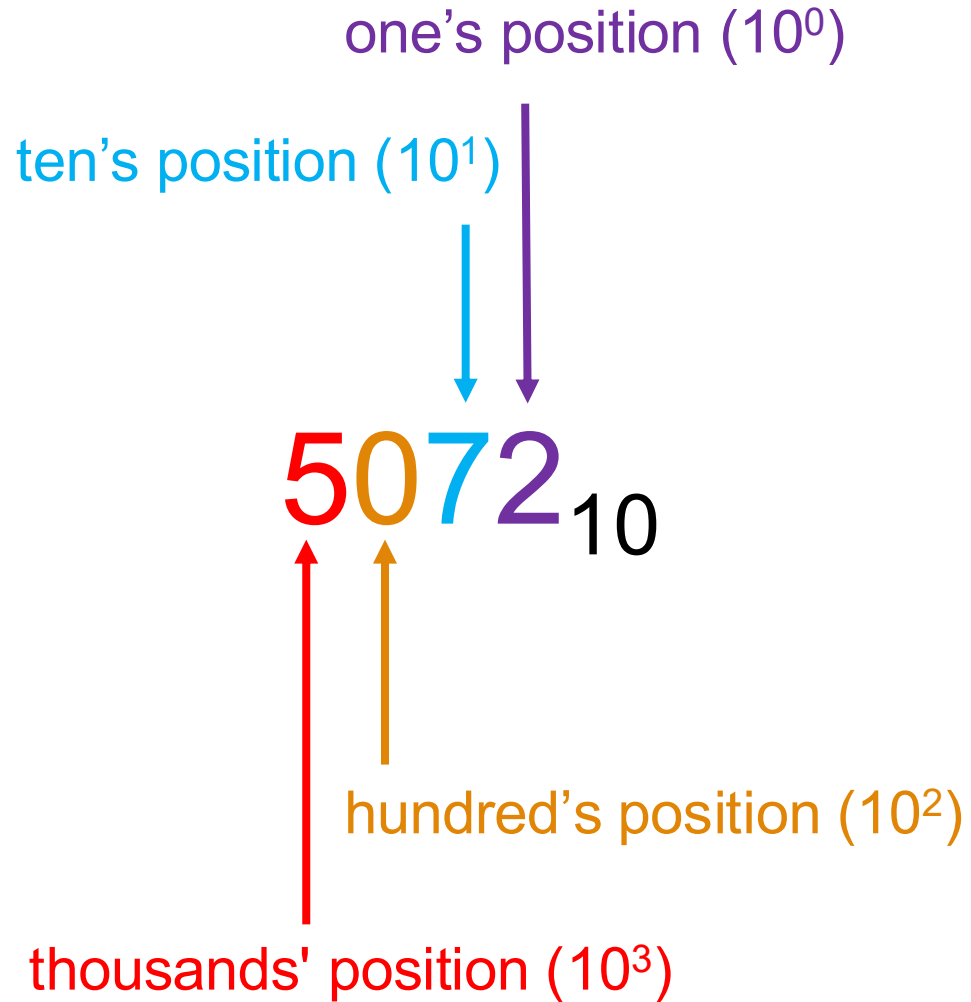
Decimal Notation

The diagram illustrates the expansion of the decimal number 5072_{10} into its place value components. Arrows point from each digit of the number to its corresponding term in the sum below.

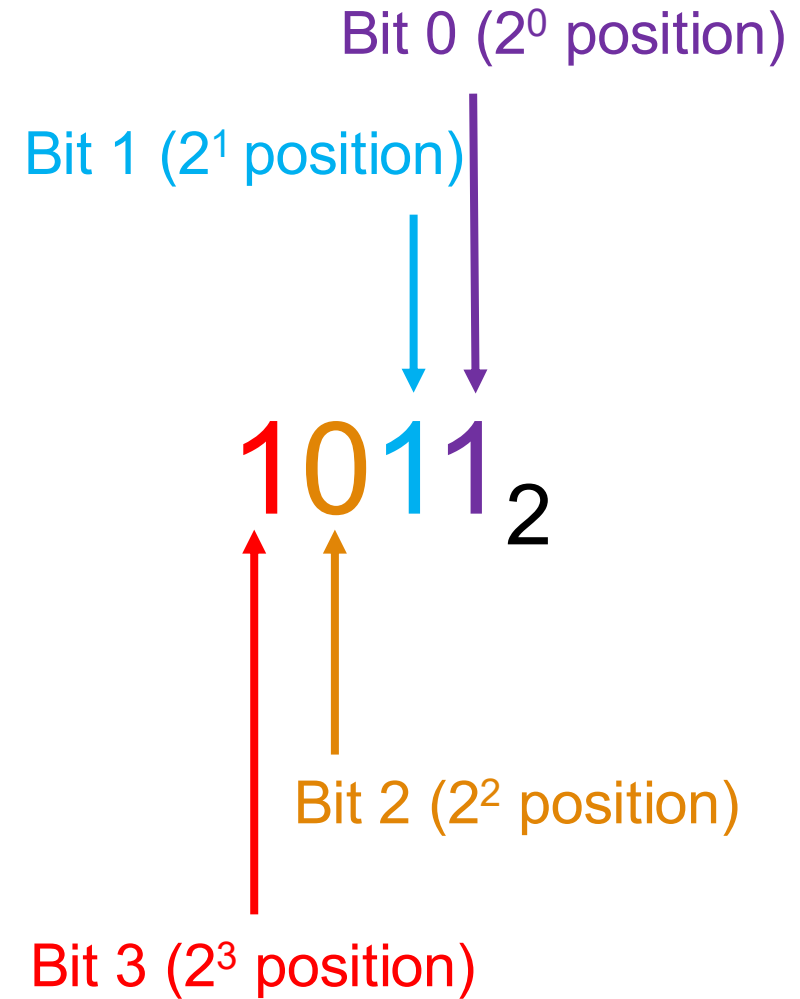
$$(5 \times 10^3) + (0 \times 10^2) + (7 \times 10^1) + (2 \times 10^0)$$
$$5000 + 0 + 70 + 2$$
$$5072_{10}$$

Base-2

Decimal



Binary



Convert from Binary to Decimal

1011_2

Convert from Binary to Decimal

$$\begin{array}{c} 1011_2 \\ \swarrow \quad \downarrow \quad \searrow \\ (1 \times 2^3) + (0 \times 2^2) + (1 \times 2^1) + (1 \times 2^0) \end{array}$$

$$8 + 0 + 2 + 1$$

$$11_{10}$$

Binary

Decimal	Binary
0	0b 0000
1	0b 0001
2	0b 0010
3	0b 0011
4	0b 0100
5	0b 0101
6	0b 0110
7	0b 0111
8	0b 1000
9	0b 1001
10	0b 1010
11	0b 1011
12	0b 1100
13	0b 1101
14	0b 1110
15	0b 1111

You should know the
binary
representation of the
numbers 0 through
15 by heart <3

Decimal vs Binary

Decimal	Powers of 10
1	$10^0 = 1$
10	$10^1 = 10$
100	$10^2 = 100$
1000	$10^3 = 1000$

Binary	Powers of 2
0b 0001	$2^0 = 1$
0b 0010	$2^1 = 2$
0b 0100	$2^2 = 4$
0b 1000	$2^3 = 8$

Binary

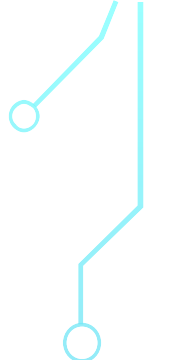
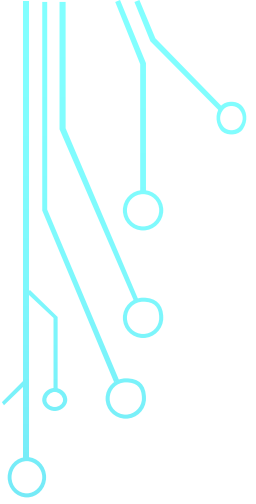
- There are two ways to indicate that a number is binary
 - Prepend 0b
 - Ex: 0b1001
 - Subscript 2
 - Ex: 1001₂
- Place binary numbers in groups of four to make it easier to read
 - Normally, spaces are used when handwritten
 - Ex: 0b1010 1111 0101
 - And underscores are used for typed values
 - Ex: 0b1010_1111_0101
 - If you cannot evenly break up the number into groups of four, start making the groups from the right
 - Ex: 0b10_1010_0011

Binary Bit Positions

- We'll often want to reference individual digits of a binary number
- To do this, we need to give a name to each bit
- The bits are named $n-1$ to 0 where n is the number of bits in the number

1 0 1 1₂

Bit 3 Bit 2 Bit 1 Bit 0



Converting from Decimal to Binary

Powers of 2

Decimal	Binary	Powers of 2
0	0b 0000	
1	0b 0001	2^0
2	0b 0010	2^1
3	0b 0011	
4	0b 0100	2^2
5	0b 0101	
6	0b 0110	
7	0b 0111	
8	0b 1000	2^3
9	0b 1001	
10	0b 1010	
11	0b 1011	
12	0b 1100	
13	0b 1101	
14	0b 1110	
15	0b 1111	

Powers of 2

$$3 = 2^1 + 2^0 = 0b0011$$

There is a 1 in
the 2^1 and 2^0
positions.

Decimal	Binary	Powers of 2
0	0b 0000	
1	0b 0001	2^0
2	0b 0010	2^1
3	0b 0011	$2^1 + 2^0$
4	0b 0100	2^2
5	0b 0101	
6	0b 0110	
7	0b 0111	
8	0b 1000	2^3
9	0b 1001	
10	0b 1010	
11	0b 1011	
12	0b 1100	
13	0b 1101	
14	0b 1110	
15	0b 1111	

All other numbers
can be formed by
adding powers of 2.

Powers of 2

$$5 = 2^2 + 2^0 = 0b0101$$

There is a 1 in
the 2^2 and 2^0
positions.

Decimal	Binary	Powers of 2
0	0b 0000	
1	0b 0001	2^0
2	0b 0010	2^1
3	0b 0011	$2^1 + 2^0$
4	0b 0100	2^2
5	0b 0101	$2^2 + 2^0$
6	0b 0110	
7	0b 0111	
8	0b 1000	2^3
9	0b 1001	
10	0b 1010	
11	0b 1011	
12	0b 1100	
13	0b 1101	
14	0b 1110	
15	0b 1111	

All other numbers
can be formed by
adding powers of 2.

Powers of 2

$$7 = 2^2 + 2^1 + 2^0 \\ = 0b0111$$

There is a 1 in
the 2^2 , 2^1 , and 2^0
positions.

Decimal	Binary	Powers of 2
0	0b 0000	
1	0b 0001	2^0
2	0b 0010	2^1
3	0b 0011	$2^1 + 2^0$
4	0b 0100	2^2
5	0b 0101	$2^2 + 2^0$
6	0b 0110	$2^2 + 2^1$
7	0b 0111	$2^2 + 2^1 + 2^0$
8	0b 1000	2^3
9	0b 1001	
10	0b 1010	
11	0b 1011	
12	0b 1100	
13	0b 1101	
14	0b 1110	
15	0b 1111	

All other numbers
can be formed by
adding powers of 2.

Powers of 2

Decimal	Binary	Powers of 2
0	0b 0000	
1	0b 0001	2^0
2	0b 0010	2^1
3	0b 0011	$2^1 + 2^0$
4	0b 0100	2^2
5	0b 0101	$2^2 + 2^0$
6	0b 0110	$2^2 + 2^1$
7	0b 0111	$2^2 + 2^1 + 2^0$
8	0b 1000	2^3
9	0b 1001	$2^3 + 2^0$
10	0b 1010	$2^3 + 2^1$
11	0b 1011	$2^3 + 2^1 + 2^0$
12	0b 1100	$2^3 + 2^2$
13	0b 1101	$2^3 + 2^2 + 2^0$
14	0b 1110	$2^3 + 2^2 + 2^1$
15	0b 1111	$2^3 + 2^2 + 2^1 + 2^0$

Convert from Decimal to Binary

- To convert a decimal number to binary, identify the combination of powers of 2 that sum to the given number.

Convert from Decimal to Binary

Ex: convert 114 to binary

1. Determine the largest power of 2 that is less than or equal to the current number

largest power of 2 less than 114 is 64

Convert from Decimal to Binary

Ex: convert 114 to binary

1. Determine the largest power of 2 that is less than or equal to the current number
2. Place a 1 in the corresponding bit position of the final answer

largest power of 2 less than 114 is 64

0b						
	64	32	16	8	4	2
	1					

Convert from Decimal to Binary

Ex: convert 114 to binary

1. Determine the largest power of 2 that is less than or equal to the current number
2. Place a 1 in the corresponding bit position of the final answer

largest power of 2 less than 114 is 64

0b	1						
	64	32	16	8	4	2	1

Convert from Decimal to Binary

Ex: convert 114 to binary

1. Determine the largest power of 2 that is less than or equal to the current number
2. Place a 1 in the corresponding bit position of the final answer
3. Subtract the power of two from the current number

largest power of 2 less than 114 is 64

0b	1						
	64	32	16	8	4	2	1

$$114 - 64 = 50$$

Convert from Decimal to Binary

Ex: convert 114 to binary

1. Determine the largest power of 2 that is less than or equal to the current number
2. Place a 1 in the corresponding bit position of the final answer
3. Subtract the power of two from the current number
4. Repeat until the current number is 0

largest power of 2 less than 114 is 64

0b	1						
	64	32	16	8	4	2	1

$$114 - 64 = 50$$

Convert from Decimal to Binary

Ex: convert 114 to binary

$$\begin{array}{r} 114 \\ - 64 \\ \hline 50 \end{array}$$

0b	1						
	64	32	16	8	4	2	1

Convert from Decimal to Binary

Ex: convert 114 to binary

$$\begin{array}{r} 114 \\ - 64 \\ \hline 50 \\ - 32 \\ \hline 18 \\ - 16 \\ \hline 2 \\ - 2 \\ \hline 0 \end{array}$$

0b	1	1	1	0	0	1	0
	64	32	16	8	4	2	1

Active Learning: Convert between Decimal and Binary

Q1. 0b 0010 0011

Q2. 0b 0101 0100

Q3. 98

Q4. 132

Hexadecimal (Base-16)

Hexadecimal (Hex)

- Method of representing binary that is easier for humans to read
- Base of 16
 - There are 16 digits ranging from 0 – F
- Denoted by prepending a *0x* or subscripting a *16*

You should know the
hex and binary
representation of the
numbers 0 through 15
by heart <3

Decimal	Binary	Hexadecimal
0	0b 0000	0x0
1	0b 0001	0x1
2	0b 0010	0x2
3	0b 0011	0x3
4	0b 0100	0x4
5	0b 0101	0x5
6	0b 0110	0x6
7	0b 0111	0x7
8	0b 1000	0x8
9	0b 1001	0x9
10	0b 1010	0xA
11	0b 1011	0xB
12	0b 1100	0xC
13	0b 1101	0xD
14	0b 1110	0xE
15	0b 1111	0xF

Hexadecimal (Hex)

How do you think you represent the decimal number 16 in hexadecimal notation?

Decimal	Binary	Hexadecimal
0	0b 0000	0x0
1	0b 0001	0x1
2	0b 0010	0x2
3	0b 0011	0x3
4	0b 0100	0x4
5	0b 0101	0x5
6	0b 0110	0x6
7	0b 0111	0x7
8	0b 1000	0x8
9	0b 1001	0x9
10	0b 1010	0xA
11	0b 1011	0xB
12	0b 1100	0xC
13	0b 1101	0xD
14	0b 1110	0xE
15	0b 1111	0xF

Convert between Binary and Hex

Convert from Binary to Hex

Ex: Convert 0b111010 to hex

Convert from Binary to Hex

1. Group the digits into groups of 4 from right to left

Ex: Convert 0b111010 to hex

0b 11 1010

Convert from Binary to Hex

1. Group the digits into groups of 4 from right to left
2. Prepend any leading zeros needed to make the leftmost group have four binary digits

Ex: Convert 0b111010 to hex

0b 11 1010

0b 0011 1010

Convert from Binary to Hex

1. Group the digits into groups of 4 from right to left
2. Prepend any leading zeros needed to make the leftmost group have four binary digits
3. Convert each group to the corresponding hex character

Ex: Convert 0b111010 to hex

0b 11 1010

0b 0011 1010

0x 3A

Convert from Hex to Binary

Ex: Convert 0x3F54 to binary

Convert from Hex to Binary

1. Convert each hex digit to its corresponding binary digits

Ex: Convert 0x3F54 to binary

0b 0011 1111 0101 0100

Convert from Hex to Binary

1. Convert each hex digit to its corresponding binary digits
2. You can optionally drop the leading zeros

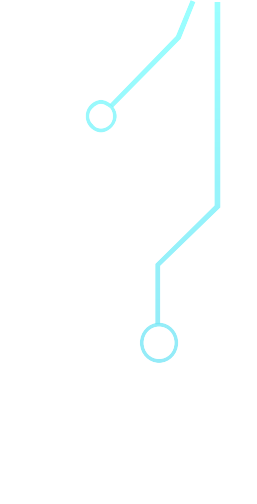
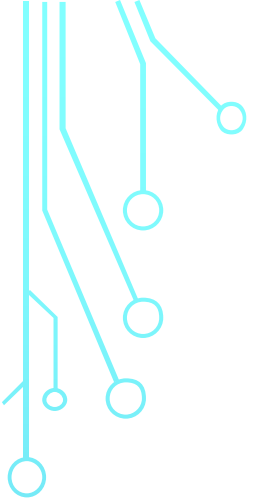
Ex: Convert 0x3F54 to binary

0b 0011 1111 0101 0100

0b 11 1111 0101 0100

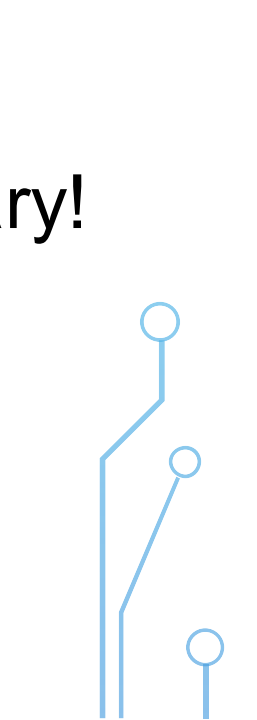
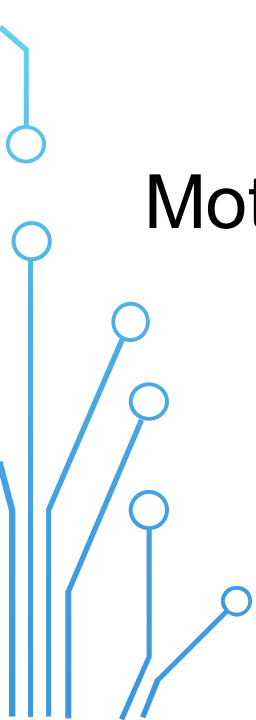
Unsigned Numbers

What range of unsigned numbers can you represent with n bits?



One's Complement

Motivation: Provides a way to represent negative numbers in binary!



One's Complement: Positive Values

- Positive values are represented the same as unsigned binary
 - Except that all positive values must begin with a leading 0.

Decimal Value	Unsigned Representation (with fewest # of bits)	One's Complement Representation (with fewest # of bits)
5	0b101	
10	0b1010	
15	0b1111	
211	0b1101_0011	

One's Complement: Positive Values

- Positive values are represented the same as unsigned binary
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Decimal Value	Unsigned Representation (with fewest # of bits)	One's Complement Representation (with fewest # of bits)
5	0b101	0b0101
10	0b1010	0b0_1010
15	0b1111	0b0_1111
211	0b1101_0011	0b0_1101_0011

Negative Values in One's Complement

- To represent a negative number, start with its positive counterpart and flip all the bits.

Decimal Value	One's Complement Representation (with fewest # of bits)
5	0b0101
10	0b0_1010
15	0b0_1111
211	0b0_1101_0011

Decimal Value	One's Complement Representation (with fewest # of bits)
-5	
-10	
-15	
-211	

All positive values have an msb of 0.

Negative Values in One's Complement

- To represent a negative number, start with its positive counterpart and flip all the bits.

Decimal Value	One's Complement Representation (with fewest # of bits)
5	0b0101
10	0b0_1010
15	0b0_1111
211	0b0_1101_0011

All positive values have an msb of 0.

Decimal Value	One's Complement Representation (with fewest # of bits)
-5	0b1010
-10	0b1_0101
-15	0b1_0000
-211	0b1_0010_1100

All negative values have an msb of 1.

One's Complement Examples

Represent -5 in 7-bit one's complement binary.

1. Write 5 in unsigned notation using 7 bits
2. Flip the bits

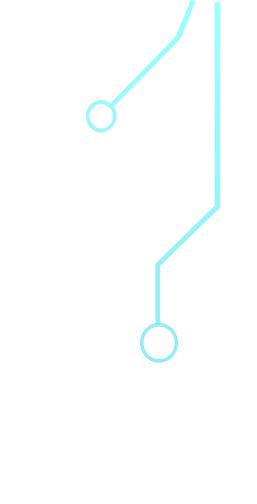
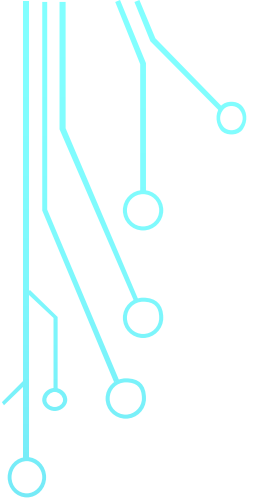
Represent 4 in 6-bit one's complement binary.

1. Write 4 in unsigned notation using 6 bits

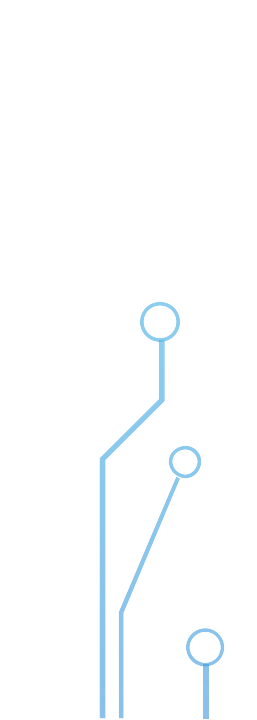
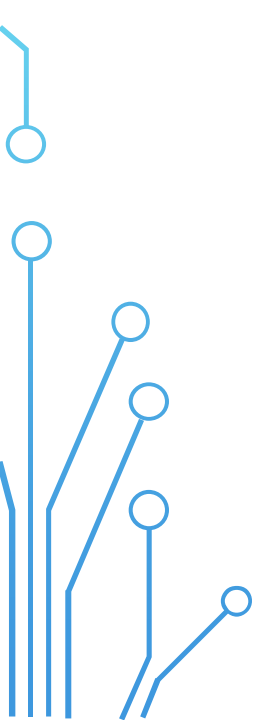
We don't flip the bits because this is a positive value!

One's Complement Shortcomings

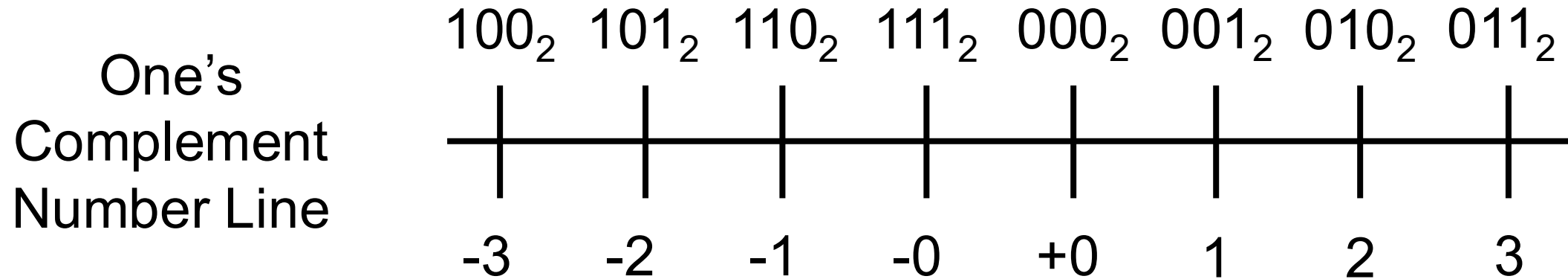
- In one's complement, zero has two representations:
 - all zeros (000...0)
 - all ones (111...1).
- This is a drawback because it wastes a possible bit pattern that could have been used to represent an additional unique value.



Two's Complement

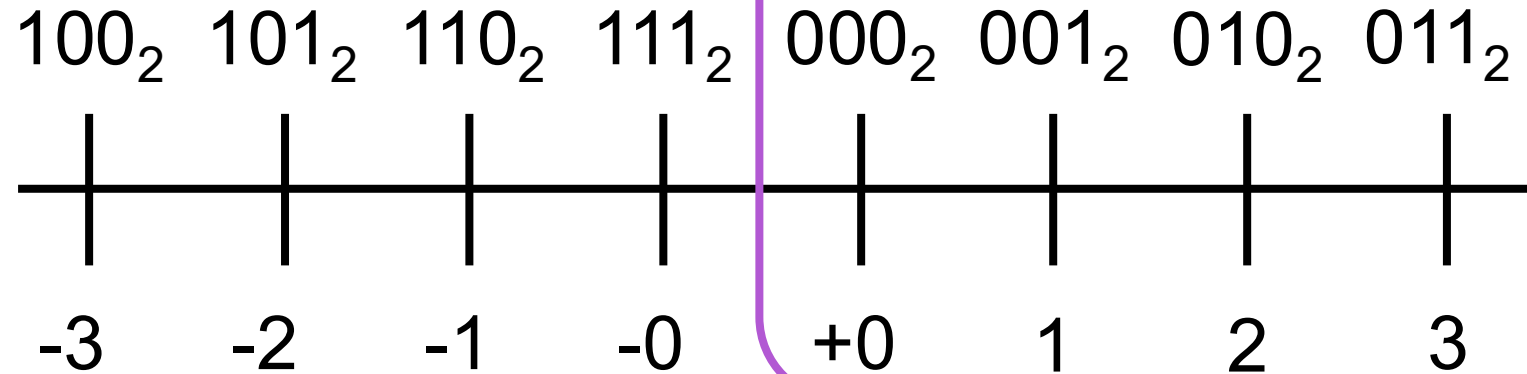


One's Complement vs Two's Complement Number Line

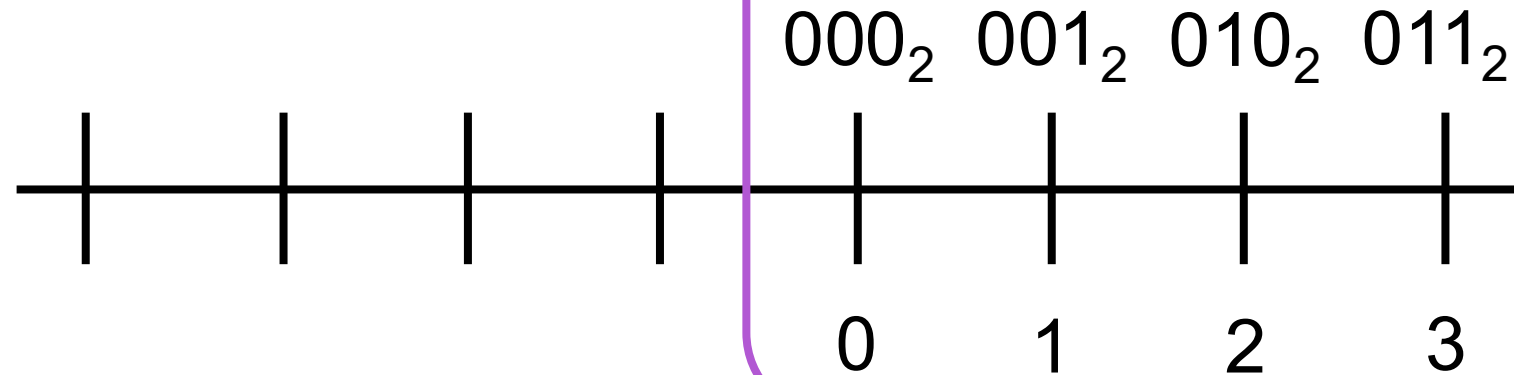


One's Complement vs Two's Complement Number Line

One's
Complement
Number Line



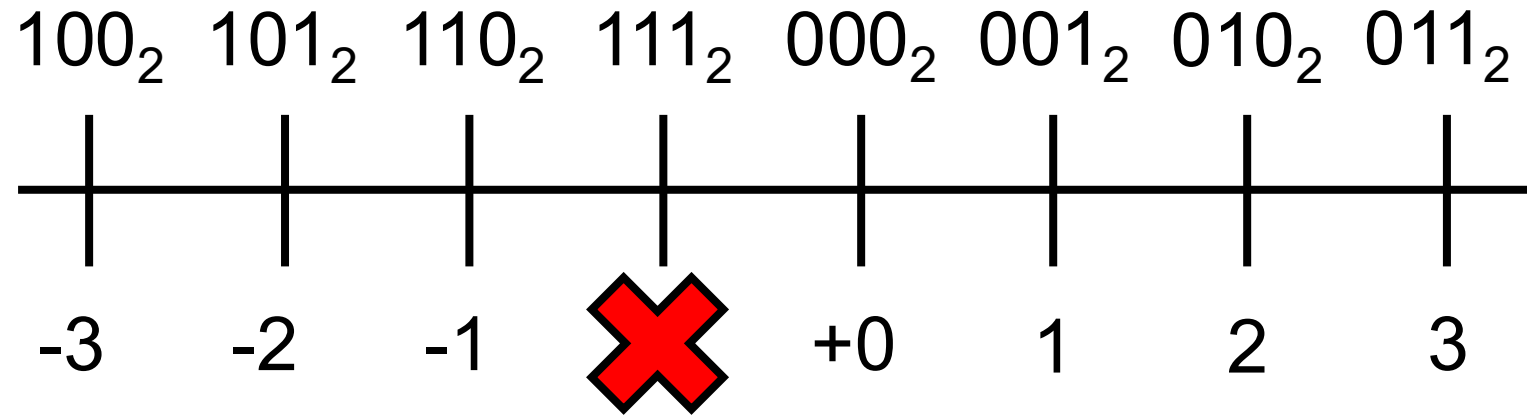
Two's
Complement
Number Line



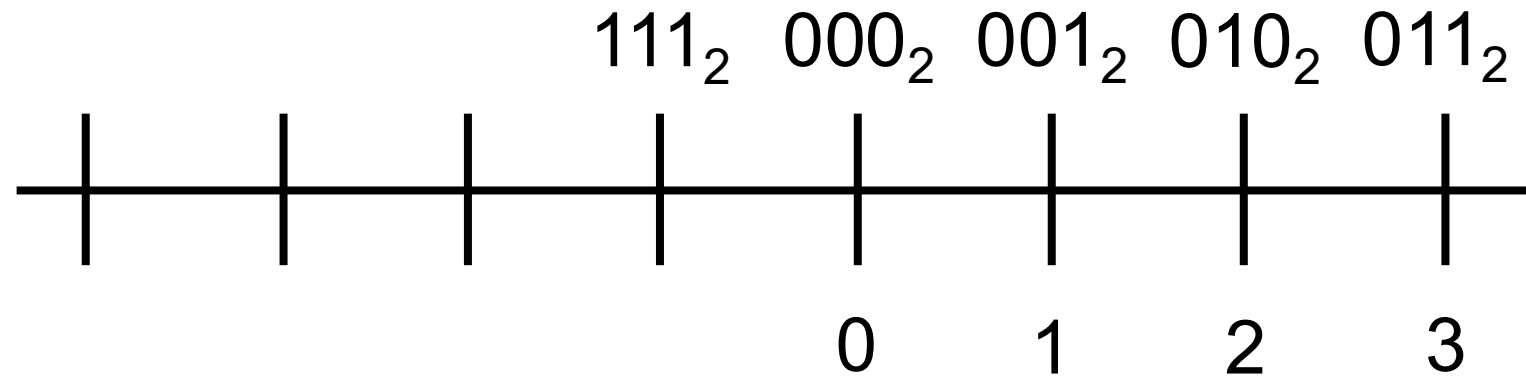
same

One's Complement vs Two's Complement Number Line

One's
Complement
Number Line

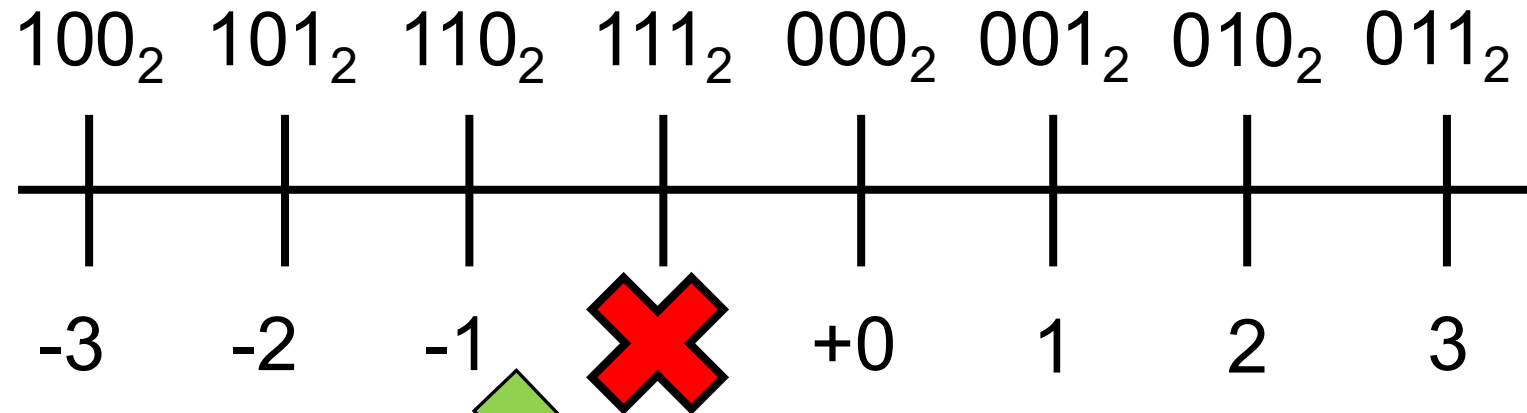


Two's
Complement
Number Line

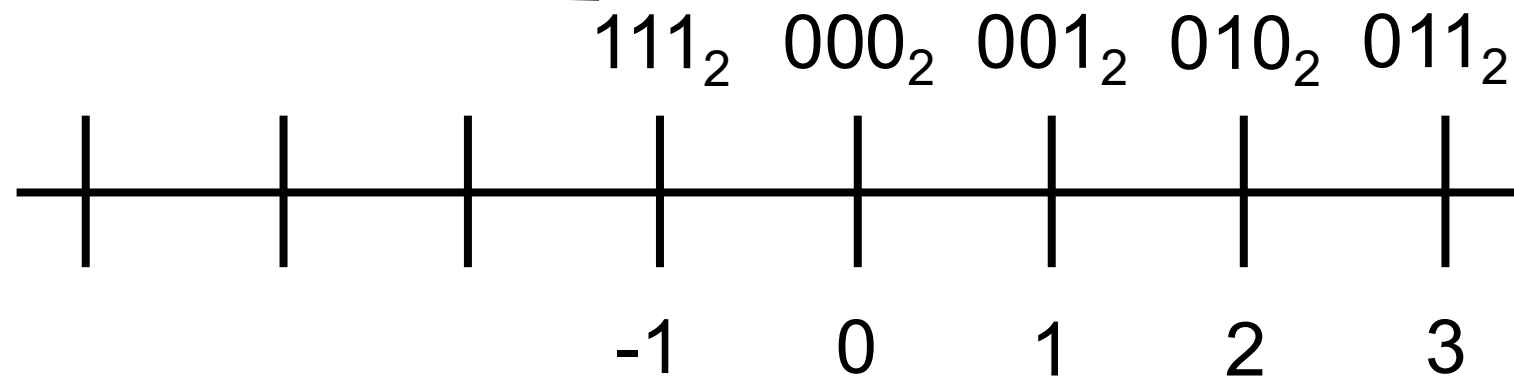


One's Complement vs Two's Complement Number Line

One's
Complement
Number Line

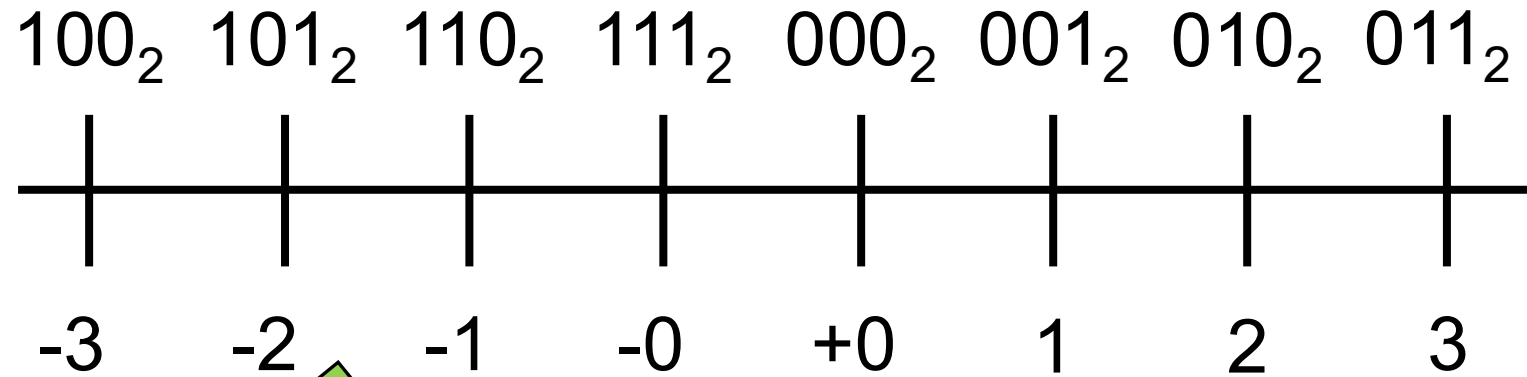


Two's
Complement
Number Line

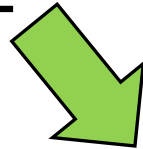
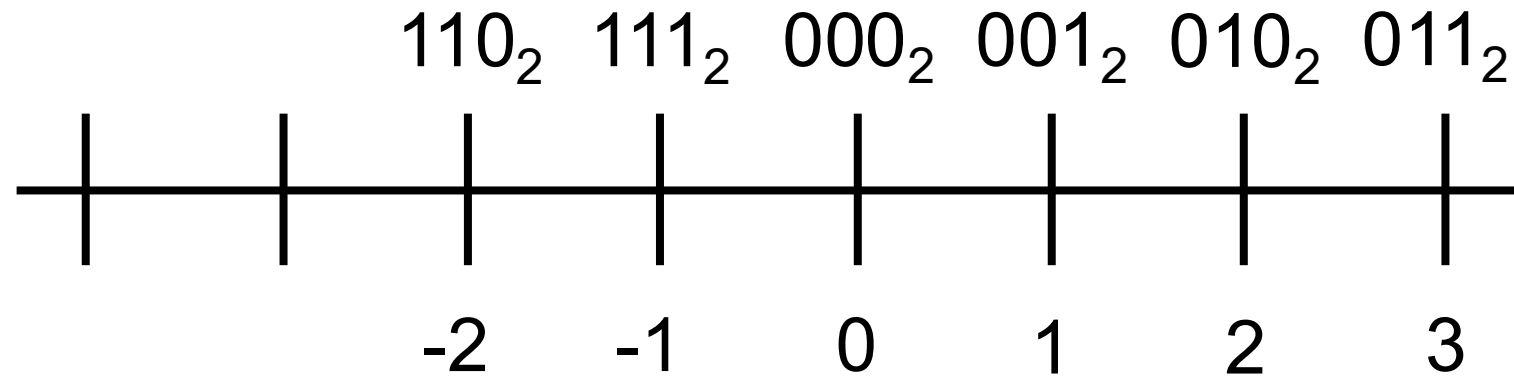


One's Complement vs Two's Complement Number Line

One's
Complement
Number Line

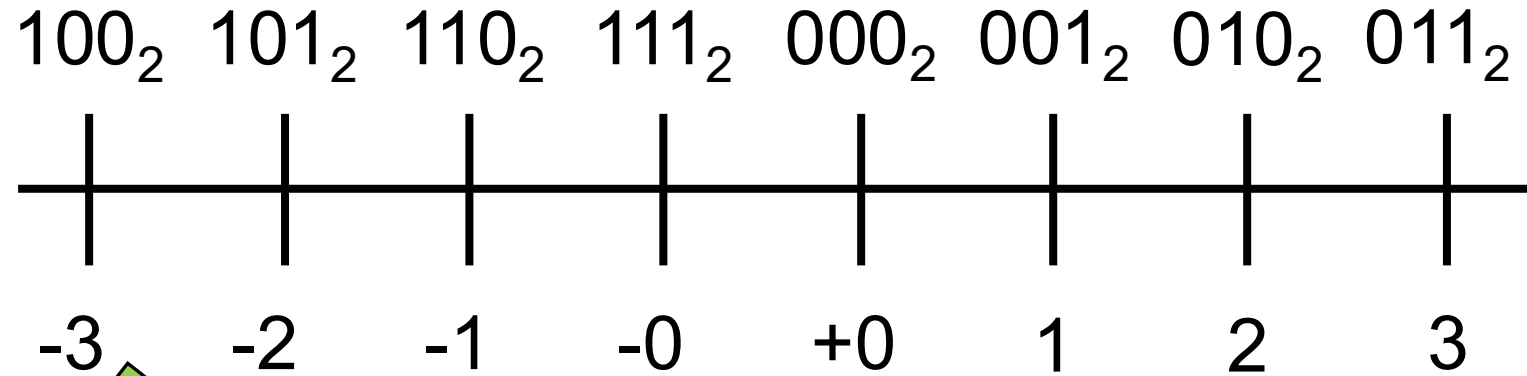


Two's
Complement
Number Line

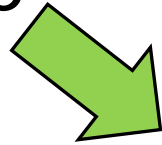
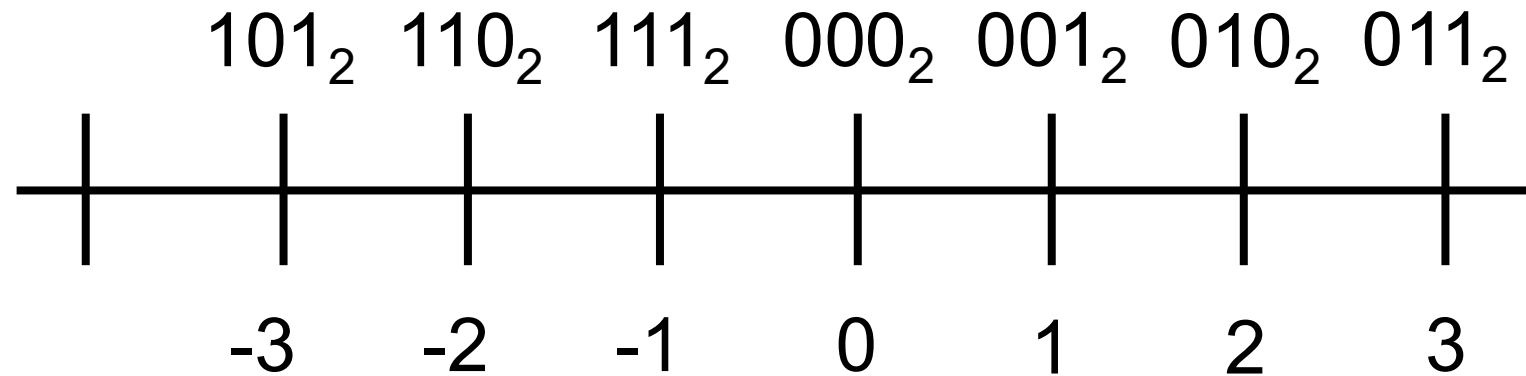


One's Complement vs Two's Complement Number Line

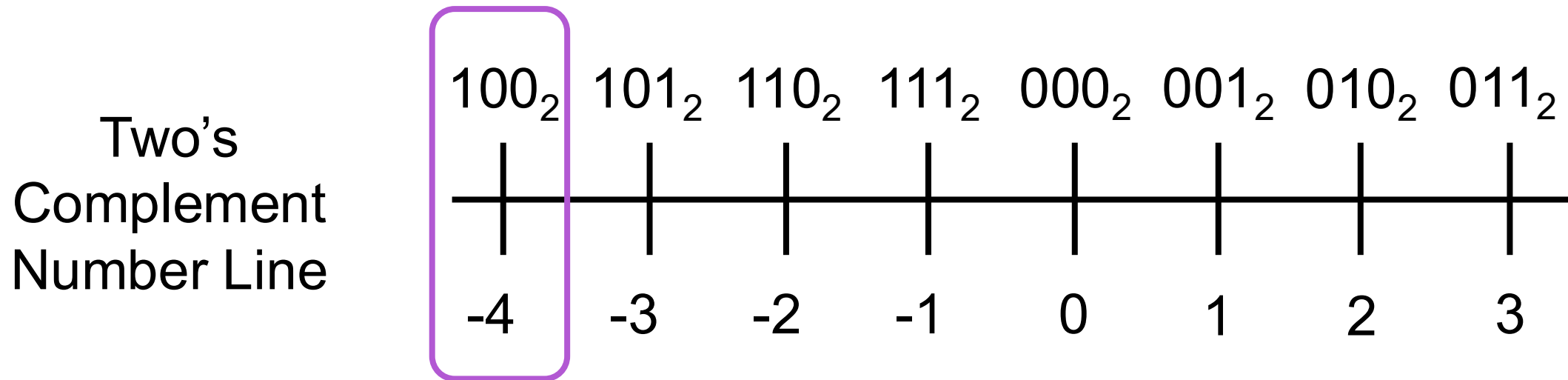
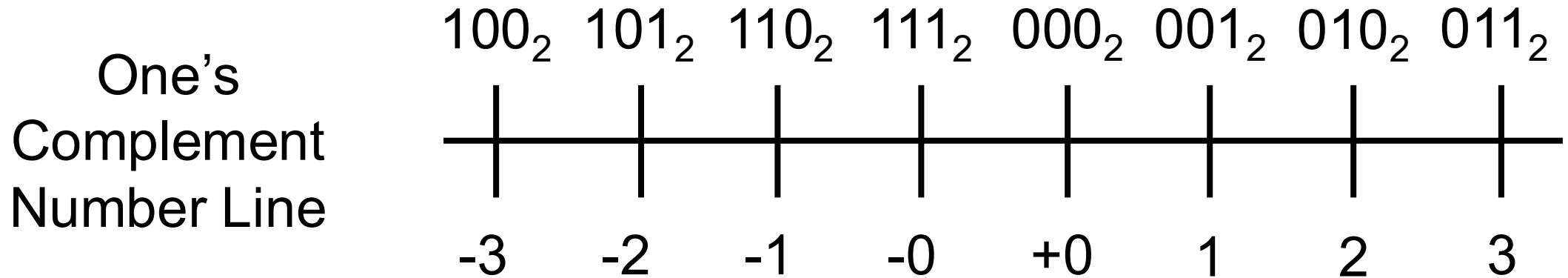
One's
Complement
Number Line



Two's
Complement
Number Line



One's Complement vs Two's Complement Number Line



Now, we can represent an additional value!

Decimal Two's Complement Representation

Step 1. Find the one's complement representation

Step 2. If negative, add one

Example: Decimal Two's Complement Representation

- Express -7 in 6-bit two's complement form.

Step 1. Find the one's complement form

Step 2. If negative, add one

Example: Decimal Two's Complement Representation

- Express -7 in 6-bit two's complement form.

Step 1. Find the one's complement

- One's complement form of +7: 0b00 0111
- One's complement form of -7: 0b11 1000

Step 2. If negative, add one

- 0b11_1001

Two's complement decimal

- **Step 1:** Check the **Most Significant Bit (MSB)**

If the MSB is 0, the number is positive.

- **Step 2:** You can interpret it the same way as an unsigned binary number.

If the MSB is 1, the number is negative.

- **Step 2:** Flip the bits (invert all bits: change 0 to 1 and 1 to 0).
- **Step 3:** Add 1 to the flipped bits.
- **Step 4:** The original number is the negative of this result.

Example: Two's complement decimal

- Convert the following two's complement binary number to decimal: 0b1101.
- **Step 1:** The MSB is 1, so the number is negative.

- **Step 2:** Flip the bits (invert all bits: change 0 to 1 and 1 to 0).
 - 0b0010
- **Step 3:** Add 1 to the flipped bits.
 - 0b0011
- **Step 4:** The original number is the negative of this result.
 - The number in step 3 is 3, so the answer is -3

Two's Complement

- We only have one zero now!
- The **most significant bit (msb)** represents the sign of the value
 - 0 means its positive
 - 1 means its negative

Two's Complement – your turn!

Q1: Express -15 in 7-bit two's complement form.

Q2: What is the minimum number of bits required to represent 20 in two's complement?

Q3: Convert the following two's complement binary number to decimal: 0b11111110.

Q4: Convert the following two's complement binary number to decimal: 0b 0011 1000

Representations Summary

- **Remember, bits are just bits!** We can't determine their value until we know the representation used to encode them.
- Representations we have learned so far
 - Unsigned
 - One's Complement
 - Two's Complement

Representations – your turn!

What is the value of the following binary numbers interpreted in unsigned, one's complement, and two's complement notation?

Q1. 1111_2

unsigned:

1's complement:

2's complement:

Q2. 0101_2

unsigned:

1's complement:

2's complement:

Q3. 1010_2

unsigned:

1's complement:

2's complement: